

KUNSAN AB, Korea, Republic of

WMO: 471410

Lat: 35.904N

Lon: 126.616E

Elev: 10

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43219

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.2	-13.2	1.2	-4.6	-11.6	1.4	-3.5	14.6	-1.1	12.6	-1.0	3.4	100	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	32.1	26.4	31.0	26.2	29.9	25.7	27.4	29.9	26.9	29.4	26.3	28.8	4.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	28.7	26.2	21.6	28.2	25.8	21.2	28.0	87.2	30.0	84.8	29.8	82.5	28.9	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.7	8.4	-10.6	34.3	2.2	1.3	-12.3	35.2	-13.6	36.0	-14.8	36.7	-16.4	37.7		
			-11.4	28.7	2.1	0.9	-12.9	29.3	-14.1	29.9	-15.2	30.4	-16.7	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	0.0	1.8	6.0	11.6	17.2	21.8	25.8	22.1	15.9	9.0	2.5
		DBStd	9.64	3.53	3.30	3.11	3.32	2.55	1.92	2.19	2.27	2.50	3.00	3.72
		HDD10.0	985	309	231	129	19	0	0	0	0	2	62	233
		HDD18.3	2526	567	463	382	202	53	1	0	2	86	279	491
		CDD10.0	2239	0	1	6	69	222	353	489	520	362	184	33
		CDD18.3	738	0	0	0	1	17	104	231	262	114	9	0
		CDH23.3	5422	0	0	0	4	55	337	1872	2557	574	21	1
		CDH26.7	1481	0	0	0	1	3	33	494	859	91	1	0
(14)	Wind	WSAvg	3.6	3.8	4.1	4.4	4.0	3.5	3.2	3.5	3.4	3.1	3.3	3.7
(15)	Precipitation	PrecAvg	1273	33	38	52	83	86	147	278	270	137	53	54
(16)		PrecMax	1797	80	101	110	243	149	338	571	573	585	138	174
(17)		PrecMin	836	8	4	12	25	22	13	47	51	19	7	11
(18)		PrecStd	258	17	25	25	50	36	87	126	141	122	41	38
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	11.0	14.8	18.0	23.2	26.8	29.2	33.6	34.8	30.8	25.2	20.8	14.0
(20)		0.4% MCWB	8.2	12.0	13.0	16.4	18.6	22.2	26.4	26.5	24.2	18.7	16.2	10.4
(21)		2% DB	8.2	10.9	15.1	20.8	24.9	27.5	31.8	32.9	28.9	23.8	18.1	11.2
(22)		2% MCWB	5.8	7.6	11.4	15.0	18.3	21.9	26.4	26.5	23.5	18.7	14.9	8.4
(23)		5% DB	6.8	8.3	13.0	18.9	23.2	26.2	30.2	31.8	27.6	22.1	16.7	9.8
(24)		5% MCWB	4.6	5.8	9.6	14.4	17.8	21.7	26.1	26.4	22.7	17.6	13.9	7.5
(25)		10% DB	5.1	6.9	11.1	17.1	21.9	25.2	29.2	30.8	26.2	21.0	15.0	8.1
(26)		10% MCWB	3.1	4.8	8.2	13.3	17.3	21.4	25.7	26.2	21.9	16.8	12.1	6.1
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	8.8	12.7	14.9	18.9	22.1	25.5	28.2	28.3	26.6	22.1	17.8	11.6
(28)		0.4% MCDB	9.9	13.8	16.2	21.3	24.3	26.8	29.9	30.8	28.7	23.7	19.2	13.2
(29)		2% WB	6.2	8.2	12.4	16.8	20.2	24.1	27.2	27.7	25.3	20.2	15.9	9.2
(30)		2% MCDB	7.6	9.9	14.1	19.0	22.5	25.4	29.5	30.4	27.2	22.1	17.1	10.7
(31)		5% WB	4.7	6.3	10.2	15.2	19.1	23.2	26.7	27.1	24.2	18.7	14.3	7.7
(32)		5% MCDB	6.1	7.8	12.0	17.6	21.8	24.6	29.1	29.9	26.1	20.7	16.0	9.2
(33)		10% WB	3.4	4.9	8.7	13.9	18.1	22.4	26.2	26.6	23.1	17.6	12.8	6.2
(34)		10% MCDB	4.8	6.4	10.7	16.1	20.7	24.0	28.4	29.3	25.4	19.9	14.6	7.8
(35)	Mean Daily Temperature Range	MDBR	7.2	7.7	8.2	8.4	7.9	6.5	5.4	6.2	7.5	8.7	8.2	7.5
(36)		5% DB MCDBR	8.6	10.1	11.0	10.8	10.2	7.8	6.6	7.5	8.4	9.6	9.5	9.2
(37)		5% WB MCWBR	6.6	7.4	7.5	6.4	5.3	4.0	3.1	3.1	4.3	5.6	7.1	7.0
(38)		5% WB MCDBR	7.9	8.9	9.9	9.0	8.4	6.2	5.8	6.3	6.5	8.0	8.5	8.5
(39)	Clear-Sky Solar Irradiance	taub	0.455	0.494	0.568	0.589	0.608	0.679	0.630	0.572	0.502	0.486	0.472	0.455
(40)		taud	1.923	1.861	1.721	1.720	1.736	1.663	1.819	1.963	2.078	1.982	1.959	1.935
(41)		Ebn at Noon	691	719	703	719	713	663	693	726	756	719	670	658
(42)		Edh at Noon	147	175	220	231	231	248	211	179	152	153	141	137
(44)	All-Sky Solar Radiation	RadAvg	2.35	3.19	4.37	5.08	5.59	5.11	4.47	4.61	4.20	3.67	2.50	2.05
(45)		RadStd	0.17	0.29	0.41	0.40	0.57	0.47	0.62	0.67	0.42	0.29	0.28	0.16

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (43 neighbors)	N/A	N/A	-0.76	N/A	N/A	N/A	N/A	N/A	+72	+43	

Nomenclature: See separate page